Roots & Reptiles

Plants and reptiles (plus frogs and salamanders) live all around North Carolina. You will learn to name them, know which ones are NC state symbols, and explain how they grow, what they need, and why they matter.

My name: _____

My partner: _____

Our meetings

Lesson	Date	What we'll do
Lesson 1	Fri Oct 9	Meet the Event: Plants, Herps, and NC Symbols
Lesson 2	Fri Nov 13	Tree Detectives: Leaves, Bark, and Seeds
Lesson 3	Fri Jan 8	Plant Parts, Life Cycles, and Leaves of Three
Lesson 4	Fri Feb 5	Herps of North Carolina
Lesson 5	Fri Mar 5	Mock Tournament
Lab I1	Before Lesson 2	Leaf Hunt: Start Your Tree Pages (home or an open Friday)
Lab I2	Before Lesson 4	Bean Tropism Lab and Plant Doctor (home or an open Friday)

At the tournament

Partners rotate through timed stations for up to 50 minutes (no going back), identifying plants, reptiles, and amphibians from the official list and answering questions, using up to 2 hand lenses and 2 references (field guides or 1-inch binders they built).

Keep every handout for this event behind this divider, in order. Bring this binder to every Field B Fridays meeting.

Roots & Reptiles: What I can do

By tournament day, you should be able to do everything on this list. Check **Practicing** when you've started, and **Got it!** when you can do it without help. The lesson column tells you which handout to look back at: L = a group meeting, I = an independent lab you do at home or on an open Friday.

What I can do	Lesson	Practicing	Got it!
I can name every NC state symbol on the specimen list: pine trees, Carolina lily, scuppernong grape, Venus flytrap, eastern box turtle, Pine Barrens treefrog, and marbled salamander.	L1 L2 L3 L4 L5 I1	☐	☐
I can identify all 14 trees on the list by their leaves, bark, and seeds, including telling the 4 pines apart by needle bundles.	L2 L5 I1	☐	☐
I can identify English ivy, Carolina lily, poison ivy, kudzu, scuppernong grape, and Venus flytrap, and I can tell poison ivy from kudzu.	L3 L5	☐	☐
I can name the jobs of roots, stems, leaves, and seeds (seed coat, cotyledon, embryo).	L3 L5	☐	☐
I can label the parts of a flower (petal, sepal, stamen, anther, filament, pistil, stigma, style, ovary) and tell what they do.	L3 L5	☐	☐
I can put the stages of a plant life cycle in order: seed, germination, seedling, adult plant, flower, pollination, fruit and seeds.	L3 I2	☐	☐
I can explain photosynthesis in my own words: leaves use sunlight, water, and carbon dioxide to make sugar (food) and give off oxygen.	L3 L5	☐	☐
I can explain and give an example of gravitropism, phototropism, thigmotropism, and hydrotropism.	L5 I2	☐	☐
I can tell what plants need to grow food and predict what happens with too little light, too much or too little water, fertilizer, or crowding.	L5 I2	☐	☐
I can tell a reptile from an amphibian using skin, eggs, babies, and breathing.	L1 L4 L5	☐	☐
I can identify the alligator, 4 lizards, 9 snakes, and 5 turtles on the list, and I can name the 3 venomous snakes.	L4 L5	☐	☐
I can identify the 4 frogs and toads and 3 salamanders on the list.	L4 L5	☐	☐
I can tell what reptiles and amphibians eat and describe how they grow and change (like egg, tadpole, froglet, frog).	L4 L5	☐	☐
I can explain why reptiles and amphibians are important to ecosystems and to people.	L4 L5	☐	☐
I can describe the habitat each group lives in and the conditions it needs (sunlight, water, temperature).	L2 L4 L5 I1 I2	☐	☐
I can find an answer in my reference binder in under one minute.	L1 L2 L5 I1	☐	☐

Meet the Event: Plants, Herps, and NC Symbols

TODAY I CAN

- I can name every NC state symbol on the specimen list: pine trees, Carolina lily, scuppernong grape, Venus flytrap, eastern box turtle, Pine Barrens treefrog, and marbled salamander.
- I can tell a reptile from an amphibian using skin, eggs, babies, and breathing.
- I can find an answer in my reference binder in under one minute.

Roots & Reptiles is about plants, reptiles, and amphibians that live in North Carolina. At the tournament you and your partner visit stations and answer questions. You may bring 2 hand lenses and 2 reference binders. This page is the first page of your binder.

Today's steps: run it yourselves (22 minutes)

You need: Station A bag (sweetgum leaf and ball, oak leaf and acorn, pine bundle), Station B photo cards (8), Station C photo cards (6), Station signs A, B, C, Sample binder, Lesson 1 answer card (Event Captain). Set the timer for each step.

1. Both 3 min. Set the timer for 3 minutes. Take turns reading the intro and the 'NC state symbols' box out loud. Set out the 3 station signs and cards.
2. Both 3 min. Station A (3 minutes): open the bag. Use a hand lens to count the needles in one pine bundle. Match each seed to its leaf. Write the needle count and your matches on a sticky note.
3. Both 4 min. Station B (4 minutes): lay each photo card next to its name in the 'NC state symbols' box. Do not unfold the answer strips yet. One card is not a state symbol. Which one?
4. Both 3 min. Station C (3 minutes): sort the 6 animal cards into a Reptile pile and an Amphibian pile. Use the 'Reptile or amphibian? Clue table' to decide.
5. Both 5 min. Now unfold every answer strip to check the names. Fix your piles. Then answer the 'Quick check' questions.
6. Both 4 min. Write your name on the Roots & Reptiles divider and put this page first. Take an Independent Lab I1 page. Partner A tells Partner B how to press a leaf between paper towels in a heavy book.

Check your work: Ask your Event Captain for the Lesson 1 answer card. Check your Station A sticky note, your Station C piles, and the Quick check. Fix any you missed in a different color.

NC state symbols on our list (memorize these!)

Pine trees	State tree. On our list: loblolly, longleaf, and shortleaf pine.
Carolina lily	State wildflower. Orange, spotted petals that curl back.
Scuppernong grape	State fruit. A bronze-green muscadine grape.
Venus flytrap	State carnivorous plant. Catches insects. Grows wild only near Wilmington.
Eastern box turtle	State reptile. Can close its shell like a box.
Pine Barrens treefrog	State frog. Green with a purple stripe on its side.
Marbled salamander	State salamander. Black with white or silver bands.

Reptile or amphibian? Clue table

Clue	Reptile	Amphibian
Skin	Dry, with scales or plates	Smooth or bumpy, moist, no scales
Eggs	Leathery shell, laid on land	Soft jelly, usually laid in water
Babies	Look like small adults	Often start as larvae (like tadpoles) and change
Breathing	Lungs	Gills when young; lungs and skin later

Quick check

1. A snake has dry scales. It is a...

◯ Reptile ◯ Amphibian

2. A salamander has moist skin and laid its eggs in a pond. It is a...

◯ Reptile ◯ Amphibian

3. Which is the NC state reptile?

◯ Copperhead ◯ Eastern box turtle ◯ American alligator

4. Which is the NC state carnivorous plant?

◯ Kudzu ◯ Poison ivy ◯ Venus flytrap

At home before next time: Do Independent Lab I1 (Leaf Hunt) before Lesson 2 on Nov 13. Start soon: fall is the best time to find fallen leaves and seeds. Do not pick up poison ivy (leaves of three)! In your Nature Detective Field Journal, check off each tree on the Tree Collector page (p. 8) and make leaf rubbings (p. 10-11). Press and name 5 leaves for the Leaf Detective badge. Museum trip (Sat Nov 7, meet 10 am at the Daily Planet Cafe): bring your journal, look for live snakes, turtles, and frogs, and fill in the Museum Day page (p. 18) with 2 you saw from our list.

Tree Detectives: Leaves, Bark, and Seeds

TODAY I CAN

- I can name every NC state symbol on the specimen list: pine trees, Carolina lily, scuppernong grape, Venus flytrap, eastern box turtle, Pine Barrens treefrog, and marbled salamander.
- I can identify all 14 trees on the list by their leaves, bark, and seeds, including telling the 4 pines apart by needle bundles.
- I can describe the habitat each group lives in and the conditions it needs (sunlight, water, temperature).
- I can find an answer in my reference binder in under one minute.

Trees are identified by their leaves, bark, and seeds. Use this chart as a binder page.

Today's steps: run it yourselves (22 minutes)

You need: Sample bags 1 and 2 (items tagged 1 to 6), Extra samples bag, Sheet protectors (20), Lesson 2 answer card (Event Captain). Set the timer for each step.

1. Both 4 min. Set the timer for 4 minutes. Lay out your pressed leaves from Lab I1. Use the 'Leaf words' box to sort them into simple, compound, lobed, toothed, and needles. No leaves? Use the extra samples bag.
2. Both 8 min. Leaf Detective: Partner A holds up an item. Both use the 'Tree ID chart' to name it. Partner B writes the name and the clue in the 'Leaf Detective' table. Switch jobs after item 3.
3. Both 3 min. Answer the 'Check yourself' questions. Use the Tree ID chart.
4. Event Captain 2 min. Unfold the answer strip on each item tag. Pairs check the Leaf Detective table and the questions.
5. Both 3 min. Each partner slides a pressed leaf from home into a sheet protector. Add a sticky note label: tree name, where found, date. No leaf? Use one from the extra samples bag.
6. Both 2 min. Put this chart and your leaf pages behind the divider. Read the at-home task together.

Check your work: Ask your Event Captain for your bag's Lesson 2 answer card. Check your Leaf Detective table and Check yourself answers. Fix any you missed in a different color.

Leaf words

Simple	One leaf blade on the stem (like magnolia)
Compound	Many small leaflets on one stem (like hickory)
Lobed	Big rounded or pointed bumps on the edge (like oak)
Toothed	Saw-like edge (like red maple or beech)
Needles	Pines have needles in bundles. Count the needles in one bundle.

Tree ID chart (14 trees on the list)

Tree	Leaf clue	Bark or seed clue
American beech	Oval, toothed, straight side veins	Smooth gray bark; small nuts in a prickly husk; dead leaves stay on in winter
Black gum	Oval, shiny, smooth edge; turns bright red early in fall	Blocky bark; small dark blue berries
Bradford (Callery) pear	Shiny, roundish, wavy edge	White, smelly flowers in early spring; invasive; branches split in storms
Crepe myrtle	Small, oval, smooth edge	Smooth bark that peels in patches; pink, purple, or white flower clusters in summer
Eastern white pine	Soft needles, 5 in a bundle	Long, thin cones
Hickory	Compound: 5 to 9 leaflets on one stem	Hard nut in a husk that splits in 4 parts

Tree	Leaf clue	Bark or seed clue
Loblolly pine *	Needles 3 in a bundle, 6 to 9 in. long	Prickly cones 3 to 6 in.; most common pine in NC
Longleaf pine *	Needles 3 in a bundle, 8 to 18 in. long	Very large cones; young trees look like a grass clump
Magnolia (southern)	Large, thick, shiny, evergreen; often rusty fuzz underneath	Huge white flowers; cone-like fruit with red seeds
Oak	Many are lobed (rounded or pointed lobes)	Acorns
Red maple	3 to 5 lobes, toothed, grow in pairs; red in fall	Winged seeds in pairs (helicopters)
River birch	Diamond shape, double-toothed edge	Papery, peeling pink-tan bark; grows near rivers
Shortleaf pine *	Needles 2 or 3 in a bundle, 3 to 5 in. long	Small cones, about 2 in.
Sweetgum	Star shape with 5 points	Spiky brown gumballs

* Pine is the NC state tree. Pines stay green all year (evergreen) and grow well in sunny spots with sandy soil.

Leaf Detective (sample bag)

#	Tree name	Clue I used
1		
2		
3		
4		
5		
6		

Check yourself

1. A pine bundle has 3 needles. Which pines could it be?

2. A pine bundle has 5 soft needles. Which pine is it?

3. Which tree has a star-shaped leaf and spiky balls?

4. Which tree has winged seeds that spin like helicopters?

5. Which trees on this chart are NC state symbols?

At home before next time: Collect 3 more leaves or seeds from trees on the chart (try for ones you do not have yet; most trees drop their leaves by early December). Press them and put them in sheet protectors with a label: name, place, date. Field Journal: make 3 bark rubbings from different trees on our list (p. 10-11) for the Bark Rubber badge. Bring your leaves and journal to Lesson 3 on Jan 8.

Plant Parts, Life Cycles, and Leaves of Three

TODAY I CAN

- I can name every NC state symbol on the specimen list: pine trees, Carolina lily, scuppernong grape, Venus flytrap, eastern box turtle, Pine Barrens treefrog, and marbled salamander.
- I can identify English ivy, Carolina lily, poison ivy, kudzu, scuppernong grape, and Venus flytrap, and I can tell poison ivy from kudzu.
- I can name the jobs of roots, stems, leaves, and seeds (seed coat, cotyledon, embryo).
- I can label the parts of a flower (petal, sepal, stamen, anther, filament, pistil, stigma, style, ovary) and tell what they do.
- I can put the stages of a plant life cycle in order: seed, germination, seedling, adult plant, flower, pollination, fruit and seeds.
- I can explain photosynthesis in my own words: leaves use sunlight, water, and carbon dioxide to make sugar (food) and give off oxygen.

Every part of a plant has a job. Today you take apart a flower and a seed, then learn the bushes, vines, and flowers on our list.

Today's steps: run it yourselves (27 minutes)

You need: Flowers (2), Cup of soaked lima beans and 2 dry beans, Paper plates and plastic knives (2 each), Paper towels, Life cycle card sets (2), Vine and flower photo cards (6), Lesson 3 answer card (Event Captain). Set the timer for each step.

1. Both 4 min. Set the timer for 4 minutes. Show your partner the leaves you pressed at home. Name each one with your Lesson 2 tree chart. Slide them into sheet protectors.
2. Both 7 min. Flower: pull off parts from the outside in: sepals, petals, stamens, then the pistil in the center. Tape one of each in the flower box and label it. Use the 'Plant parts and their jobs' box to find which part makes pollen. Pollen stains, so use a paper towel.
3. Both 4 min. Seed: rub the seed coat off a soaked bean and split it open with your fingernail or a plastic knife. Find the embryo (baby plant) and 2 cotyledons (stored food) with a hand lens. Why is it bigger than a dry bean?
4. Both 4 min. Partner A lays the life cycle cards in a circle in order. Partner B checks with the 'Plant life cycle' list. Then take turns reading the 'Photosynthesis' box out loud.
5. Both 5 min. Match the 6 photo cards to the 'Bushes, vines, and flowers' table. Put poison ivy and kudzu side by side and find 2 differences. Never touch real poison ivy.
6. Both 3 min. Answer the 'Check yourself' questions. Take an Independent Lab I2 page home.

Check your work: Ask your Event Captain for the Lesson 3 answer card. Check your life cycle order and your Check yourself answers. Fix any you missed in a different color.

Plant parts and their jobs

Roots	Hold the plant in the soil and take in water and nutrients
Stem	Holds the plant up and carries water and food
Leaves	Make food for the plant using sunlight
Seed	Holds a baby plant (embryo) and its stored food (cotyledon) inside a seed coat
Flower	Makes seeds. Petals attract pollinators; sepals protect the bud
Stamen	Male part: anther (makes pollen) on a thin filament
Pistil	Female part: stigma (sticky top catches pollen), style (tube), ovary (grows into fruit with seeds)

Tape your flower parts here and label them

sepal petal stamen pistil

Photosynthesis (how plants make food)

Photosynthesis happens in leaves. A green material called chlorophyll catches sunlight. The plant uses that energy with water (from roots) and carbon dioxide (a gas from the air) to make sugar, its food. It gives off oxygen, which we breathe.

Plant life cycle

1. Seed
2. Germination: the seed takes in water and sprouts
3. Seedling
4. Adult plant
5. Flower
6. Pollination: pollen moves to the stigma
7. Fruit and seeds form, and the cycle starts again

Bushes, vines, and flowers on our list

Plant	How to know it	Fun fact
English ivy	Evergreen vine; dark green leaves with 3 to 5 pointed lobes	Invasive: climbs and covers trees
Carolina lily *	Orange flower with spots; petals curl back	State wildflower
Eastern poison ivy	Leaves of three; hairy vine; white berries	Oil causes an itchy rash. Never touch!
Kudzu	3 large, hairy leaflets; purple flowers	Invasive vine from Asia; can grow about a foot a day
Scuppernong grape *	Vine with round, toothed leaves; bronze-green grapes	State fruit
Venus flytrap *	Leaf ends in a trap with trigger hairs	State carnivorous plant; grows wild only near Wilmington

Check yourself

1. Which flower part makes pollen?

2. Which flower part grows into the fruit?

3. What 3 things does a leaf use in photosynthesis?

4. What 2 things does it make?

5. Which part of a seed is stored food?

6. How can you tell kudzu from poison ivy?

7. Which plants on this page are state symbols?

At home before next time: Do Independent Lab I2 (Bean Tropism Lab and Plant Doctor) before Lesson 4 on Feb 5. Start your bean bag this weekend so the beans have time to sprout, and draw it every day in the Bean Bag Log in your Field Journal (p. 17) for the Sprout Scientist badge. Also get ready for reptiles and amphibians (10 minutes): fill in the Reptiles | Amphibians T-chart in your journal (p. 4) with 5 differences (use the Lesson 1 clue table or the Smithsonian 'Am I a Reptile? Am I an Amphibian?' page with an adult).

Herps of North Carolina

TODAY I CAN

- I can name every NC state symbol on the specimen list: pine trees, Carolina lily, scuppernong grape, Venus flytrap, eastern box turtle, Pine Barrens treefrog, and marbled salamander.
- I can tell a reptile from an amphibian using skin, eggs, babies, and breathing.
- I can identify the alligator, 4 lizards, 9 snakes, and 5 turtles on the list, and I can name the 3 venomous snakes.
- I can identify the 4 frogs and toads and 3 salamanders on the list.
- I can tell what reptiles and amphibians eat and describe how they grow and change (like egg, tadpole, froglet, frog).
- I can explain why reptiles and amphibians are important to ecosystems and to people.
- I can describe the habitat each group lives in and the conditions it needs (sunlight, water, temperature).

Reptiles and amphibians are called herps. Use this chart as a binder page. A * means NC state symbol.

Today's steps: run it yourselves (27 minutes)

You need: Herp photo card sets (2, 26 cards each), Group label cards (2 sets of 6), Lesson 4 answer card (Event Captain). Set the timer for each step.

1. Both 4 min. Set the timer for 4 minutes. Trade T-charts from home. Check your partner's 5 differences with the Lesson 1 clue table. Fix any mistakes.
2. Both 8 min. Herp Sort: lay out the 6 group labels. Sort all 26 photo cards under them. Then use the reptile and amphibian tables to name each card, and unfold the answer strip to check.
3. Both 5 min. Find the 3 cards marked venomous in the reptile table. Put the cottonmouth next to the banded water snake and find 2 differences. Outdoors, never touch any snake.
4. Both 5 min. On your whiteboard, draw a food chain: plant, cricket, green anole, black rat snake, hawk. Draw arrows from the food to the eater. Then draw the frog life cycle in the box.
5. Both 5 min. Answer the 'Check yourself' questions. For number 7, think: what do they eat, who eats them, and what do they tell us about clean water?

Check your work: Ask your Event Captain for the Lesson 4 answer card. Check your Check yourself answers and your group counts. Fix any you missed in a different color.

Reptiles on our list

Animal	How to know it	Eats / lives
American alligator	Huge, dark, armored; wide round snout	Fish, turtles, birds; coastal swamps and rivers
Green anole	Turns green or brown; male has pink throat fan	Insects; shrubs and fences
SE five-lined skink	Shiny, 5 stripes; young have a blue tail	Insects, spiders; woods, logs
Eastern glass lizard	Legless lizard with eyelids and ear holes	Insects, snails; sandy fields
Eastern fence lizard	Rough gray-brown scales; male has blue belly	Insects; sunny pine woods, fences
Copperhead (venomous)	Copper head; hourglass bands	Mice, frogs; woods. Live birth
Cottonmouth (venomous)	Thick, dark; white inside mouth	Fish, frogs; swamps (east NC). Live birth
E. diamondback rattlesnake (venomous)	Diamonds on back; rattle	Rabbits, rodents; longleaf pine woods. Very rare
Black rat snake	Long, shiny black; great climber	Rodents, bird eggs; woods, barns
Dekay's brownsnake	Small, brown, 2 rows of dark spots	Slugs, worms; yards
Eastern hognose snake	Upturned snout; flattens neck, plays dead	Toads; sandy areas
Eastern kingsnake	Black with white or yellow chain bands	Other snakes, even venomous ones
Banded water snake	Dark crossbands; harmless	Fish, frogs; ponds, streams
Eastern garter snake	Long stripes down the body	Worms, frogs; yards, fields. Live birth
Snapping turtle	Big head, jagged tail, small bottom shell	Almost anything; ponds
Painted turtle	Smooth dark shell; red and yellow edges	Plants, insects; ponds; basks

Animal	How to know it	Eats / lives
Spotted turtle	Black shell with yellow dots	Insects, worms; wetlands
Eastern box turtle *	High domed shell that closes like a box	Berries, mushrooms, worms; forests
Yellow-bellied slider	Yellow belly; yellow patch behind eye	Plants, insects; ponds; basks on logs

Amphibians on our list

Animal	How to know it	Eats / lives
American toad	Dry, warty skin; big bumps behind eyes	Insects, worms; yards, gardens
Pine Barrens treefrog *	Green with purple side stripe; sticky toe pads	Insects; boggy sandhills
Bullfrog	Biggest frog; call sounds like 'jug-o-rum'	Insects, fish, even snakes; ponds
Southern leopard frog	Dark spots, pointed snout	Insects; wet fields, ponds
Marbled salamander *	Black with white or silver bands	Worms, insects; forest near ponds; eggs laid in fall
Red-backed salamander	Red stripe down the back; no lungs	Tiny insects; under logs; eggs on land, no tadpole
White-spotted slimy salamander	Black with white spots; sticky slime	Insects; under logs; no lungs

Draw the frog life cycle

eggs > tadpole > froglet > adult

Check yourself

1. Name the 3 venomous snakes on our list.

2. What is the NC state reptile?

3. Which 2 amphibians are NC state symbols?

4. Which snake eats other snakes?

5. Which snake plays dead?

6. List the stages of a frog's life cycle.

7. Give 2 reasons reptiles and amphibians are important.

At home before next time: Binder check before our mock tournament: make sure your binder has (1) state symbols page, (2) Lab I1 leaf hunt page, tree chart, and pressed leaves, (3) plant parts and life cycle page, (4) Lab I2 tropism and Plant Doctor page, (5) herp chart. Add colored tabs so you can flip to each section fast. Then quiz a family member: read a clue from the herp chart and see if they can find the animal. Time how fast YOU can find 5 animals. Field Journal: spot a real herp (outside, a zoo, or a nature center) and log it on the Herp Spotter Log (p. 16) for the Herp Spotter badge, or name the 3 venomous snakes and a look-alike (p. 6) for Snake Smart.

Mock Tournament

TODAY I CAN

- Everything on my can-do checklist (the page right after the divider). Today is the time to show it.

This is a practice tournament. Work with your partner. Use only your binder and hand lens. For every ID, write the name and circle Y if it is an NC state symbol or N if it is not.

Today's steps: run it yourselves (27 minutes)

You need: Station bags 1 to 5 (tree samples, photo cards, flower diagram, bean card), Station signs (5), Lesson 5 answer card (Event Captain). Set the timer for each step.

1. Event Judge 2 min. Put the 5 station signs and bags at 5 spots. Read the intro out loud: 4 minutes per station, no going back, only binders and hand lenses. Set the timer for 4 minutes.
2. Both 8 min. Stations 1 and 2. The Event Judge calls 'One minute!' at 3 minutes and 'Rotate!' at 4, then resets the timer. Pairs write each name and circle Y or N.
3. Both 12 min. Stations 3, 4, and 5. Same timing. If you finish early, check your answers in your binder.
4. Event Captain 3 min. Hand out the answer card. Each pair scores its own sheet.
5. Both 2 min. Circle 2 things you could not find fast. Write them on the 'Fix my binder' lines.

Check your work: Use the Lesson 5 answer card from your Event Captain to score your sheet. Give 1 point for each right name and 1 for each right Y or N. Fix misses in a different color.

Station 1: Trees

#	Name	Symbol?
1		Y N
2		Y N
3		Y N
4		Y N

1. Which tree is the NC state tree?

Station 2: Plants

#	Name	Symbol?
1		Y N
2		Y N
3		Y N
4		Y N

1. Which plant here is invasive?

Station 3: Plant science

1. Name parts A, B, and C and give one job for each.

2. The bean root curved down. What tropism is this?

3. What 2 things does photosynthesis make?

Station 4: Reptiles

#	Name	Symbol?
1		Y N
2		Y N
3		Y N
4		Y N

1. Why is the box turtle a reptile and not an amphibian?

Station 5: Amphibians

#	Name	Symbol?
1		Y N
2		Y N
3		Y N
4	(life stage)	Y N

1. Why are frogs important in an ecosystem?

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2. Fix my binder: 2 things we could not find fast

At home before next time: Fix the 2 things you listed. Add a page or a tab so you can find each one in under a minute. Before the tournament, flip through your whole binder once with your partner and pack 2 hand lenses. Field Journal: say all 7 NC state symbols from memory and check them off on page 3 for the State Symbol Seeker badge.

Lab I1: Leaf Hunt: Start Your Tree Pages

THIS LAB HELPS ME

- I can name every NC state symbol on the specimen list: pine trees, Carolina lily, scuppernong grape, Venus flytrap, eastern box turtle, Pine Barrens treefrog, and marbled salamander.
- I can identify all 14 trees on the list by their leaves, bark, and seeds, including telling the 4 pines apart by needle bundles.
- I can describe the habitat each group lives in and the conditions it needs (sunlight, water, temperature).
- I can find an answer in my reference binder in under one minute.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Fall is the best time to collect leaves for your tournament binder. Scientists sort leaves by their shape and edges. In this lab you collect, sort, and press leaves that will become your first binder pages.

Steps (28 minutes)

You need: Collecting bag, Paper towels and a heavy book, Paper and a peeled crayon, Hand lens (optional). Set the timer for each step.

1. Me 3 min. Set the timer for 3 minutes. Read 'Leaf words' and 'Easy trees to find in fall' out loud. Remember: never pick up leaves of three.
2. Me 10 min. With an adult's OK, walk your yard, street, or a park. Pick up 5 fallen leaves, seeds, or pine bundles from 5 different trees. Only take things from the ground. On an open Friday, use the extra samples bag.
3. Me 6 min. Set the timer for 6 minutes. Fill in 'My leaf hunt': write the leaf word and your best guess at the tree for each one. Count the needles in any pine bundle.
4. Me 4 min. Bark rubbing: hold a sheet of paper on a tree trunk and rub it with the side of the crayon. Write the tree's name on it if you know it. Tape it in the 'Bark rubbing' box or clip it behind this page.
5. Me 5 min. Lay each leaf flat between 2 paper towels inside a heavy book. Leave them for 1 to 2 weeks. Then answer the 'Check yourself' questions.

Check your work: Check your guesses with the 'Easy trees to find in fall' box. At Lesson 2 you will check them again with the full tree chart. On an open Friday, ask your Event Captain for the Lab I1 answer card.

Leaf words

Simple	One leaf blade on the stem (like magnolia)
Compound	Many small leaflets on one stem (like hickory)
Lobed	Big rounded or pointed bumps on the edge (like oak)
Toothed	Saw-like edge (like red maple or beech)
Needles	Pines have needles in bundles. Count the needles in one bundle.

Easy trees to find in fall

Sweetgum	Star leaf with 5 points; spiky brown gumballs
Red maple	3 to 5 toothed lobes; turns red; winged seeds in pairs
Oak	Most leaves are lobed; acorns
Loblolly pine	Needles 3 in a bundle, 6 to 9 inches long; the most common pine in NC
Southern magnolia	Big, thick, shiny evergreen leaf; often rusty fuzz underneath
Hickory	Compound leaf: 5 to 9 leaflets on one stem; nut in a husk that splits in 4

My leaf hunt

#	Leaf word	My guess at the tree	Where and when I found it
1			
2			
3			
4			
5			

Bark rubbing (tape it here)

Check yourself

1. A pine bundle has 3 needles, 6 to 9 inches long. Which pine is it most likely?

2. Which tree has a star-shaped leaf with 5 points?

3. A leaf has many small leaflets on one stem. What is the leaf word?

4. Which NC state symbol is a tree?

Bring to the next meeting: Bring your pressed leaves (still in the paper towels) and this page to Lesson 2 on Nov 13. You will put them in sheet protectors there.

Lab I2: Bean Tropism Lab and Plant Doctor

THIS LAB HELPS ME

- I can put the stages of a plant life cycle in order: seed, germination, seedling, adult plant, flower, pollination, fruit and seeds.
- I can explain and give an example of gravitropism, phototropism, thigmotropism, and hydrotropism.
- I can tell what plants need to grow food and predict what happens with too little light, too much or too little water, fertilizer, or crowding.
- I can describe the habitat each group lives in and the conditions it needs (sunlight, water, temperature).

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. A tropism is a way a plant grows toward or away from something. In this lab you grow beans in a bag and see which way roots and stems grow, even when you turn them sideways. Then you play Plant Doctor.

Steps (28 minutes)

You need: Zip-top bag and paper towel, 3 dry beans, Tape. Set the timer for each step.

1. Me 5 min. Day 1: wet a paper towel, fold it, and slide it into the bag. Put 3 beans on it, about 2 cm apart. Seal the bag and tape it to a window. Write the date and your prediction in 'My prediction'. On an open Friday, set it up and take it home.
2. Me 6 min. Every 3 days, look at the beans for 2 minutes. Fill in one row of the 'Bean journal': which way did the root grow? The stem? Keep the towel damp.
3. Me 3 min. When the stems are about 5 cm (2 inches) tall, turn the bag on its side and tape it that way. After 2 days, fill in the last row of the journal.
4. Me 4 min. Set the timer for 4 minutes. Read 'The 4 tropisms' out loud. Answer the 'Name the tropism' questions.
5. Me 8 min. Set the timer for 8 minutes. Read 'What plants need'. For each patient in 'Plant Doctor', write what is wrong and how to fix it.
6. Me 2 min. Finish 'What I found': which tropism did your turned bag show?

Check your work: Check 'Name the tropism' with 'The 4 tropisms' box and 'Plant Doctor' with 'What plants need'. On an open Friday or at Lesson 4, ask your Event Captain for the Lab I2 answer card and fix any you missed in a different color.

The 4 tropisms

Gravitropism	Growth in response to gravity. Roots grow down, stems grow up.
Phototropism	Growth toward light. A houseplant bends toward the window.
Thigmotropism	Response to touch. Grape and ivy vines curl around a fence. (A Venus flytrap also closes when touched.)
Hydrotropism	Growth toward water. Roots spread toward wet soil or a leaky pipe.

My prediction

1. Which way will the roots grow? Which way will the stems grow? What will happen if I turn the bag on its side?

Bean journal

Date	What the root did	What the stem did
After turning:		

What I found

1. After I turned the bag, the root _____ and the stem _____. This is called _____.

What plants need (horticulture = gardening)

Sunlight	Energy to make food
Water	Carries nutrients; needed for photosynthesis
Soil	Holds roots and stores water and nutrients
Nutrients and fertilizer	Fertilizer adds nutrients: N (nitrogen) for green leaves, P (phosphorus) for roots and flowers, K (potassium) for overall health
Space	Crowded plants compete for light, water, and nutrients

Plant Doctor

Patient	What is wrong?	How to fix it
1. Tomato in a dark closet: tall, thin, pale		
2. Soil is always soggy; leaves yellow		
3. Leaves droop and feel crispy; soil is dry		
4. 20 bean plants in one small pot, all tiny		
5. A little fertilizer added; plant is extra green		
6. A whole box of fertilizer; leaf edges brown		

Name the tropism

1. Sunflower seedlings bend toward the sun.

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2. A bean root turned sideways curves back down.

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3. A grape tendril wraps around a wire.

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4. Tree roots crack into a leaky water pipe.

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5. Which invasive vine on our list wins the competition for sunlight by covering other plants?

Bring to the next meeting: Bring this page and your bean bag (if it still looks good) to Lesson 4 on Feb 5. Put the page in your binder after Lesson 4.